- WHILE DYNAMICS DESCRIBES WHY A FLUID ELEMENT DEFORMS, KINEMATICS PROVIDES THE MATHEMATICAL MACHINERY TO DESCRIBE THE DEFORMATION.
- LAGRANGIAN DESCRIPTION OF THROWING A BALL, $x(t) = f(x_0, t)$, $v = dx(t)/dt$, $a = d^2x(t)/dt^2$ (GIVES EXPLICIT INFORMATION ON TIME HISTORY OF PARTICLE)
- EULERIAN DESCRIPTION OF THROWING A BALL, FIELD TO DESCRIBE THE BALL (PARTICULAR LOCATION, PART. TIME)
- MATERIAL DERIVATIVE: LET F BE A PROPERTY OF THE FLOW UNDER CONSIDERATION. THIS MAY BE EXPRESSED USING THE LAGRANGIAN SYSTEM $F_L(x_0, \hat{t})$, OR IN THE EULERIAN SYSTEM $F_E(x,t)$

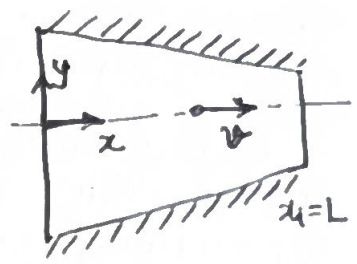
$$\Rightarrow F = F_L(x_0, \hat{t}) = F_E(x(x_0, \hat{t}), t = \hat{t})$$ ← TIME HAS TO BE EQUATED SO THAT THINGS ARE CONSISTENT.

TAKING TIME DERIVATIVE, $\frac{\partial F_L}{\partial \hat{t}} = \frac{\partial F_E}{\partial x} \cdot \frac{\partial x}{\partial \hat{t}} + \frac{\partial F_E}{\partial t} \cdot \frac{\partial t}{\partial \hat{t}}$ ← FROM CHAIN RULE OF DIFFERENTIATION

$$\Rightarrow \frac{\partial F_L}{\partial \hat{t}} = v \frac{\partial F_E}{\partial x} + \frac{\partial F_E}{\partial t}$$ ← EXPRESSION FOR MATERIAL DERIVATIVE

(ADVECTIVE COMPONENT / LOCAL COMPONENT)

- SHOW THE EXPANSION OF THE MATERIAL DERIVATIVE FOR VELOCITY, ACCELERATION.



EXAMPLE: CONSIDER A 2D, STEADY, INCOMPRESSIBLE FLOW THROUGH A PLANE CONVERGING CHANNEL. THE VELOCITY ON THE HORIZONTAL CENTERLINE IS GIVEN BY $v = v_1[1 + x/L]\hat{i}$. FIND AN EXPRESSION FOR THE ACCELERATION OF A PARTICLE MOVING ALONG THE CENTERLINE USING (A) EULERIAN APPROACH (B) LAGRANGIAN APPROACH. EVALUATE THE ACCELERATION WHEN THE PARTICLE IS AT THE BEGINNING AND AT THE END OF THE CHANNEL.

$v = v_1[1 + x/L] \Rightarrow Dv/Dt = u\,\partial v/\partial x = v_1[1 + x/L] \cdot v_1/L = v_1^2/L\,[1 + x/L]$ → EULERIAN DESC.

LAGRANGIAN, x_p: $v_1(1 + x_p/L) = dx_p/dt \Rightarrow v_1 dt = dx_p/(1 + x_p/L) \Rightarrow L\ln\left(1 + \frac{x_p}{L}\right) = v_1 t \Rightarrow x_p(t) = L(e^{v_1 t/L} - 1)$

$\Rightarrow u_p(t) = \frac{v_1}{L}\left[1 + L(e^{v_1 t/L} - 1)\right]$, $a_p(t) = v_1^2/L\, e^{v_1 t/L}$. } FOR THE PARTICLE WHICH WAS @ $x=0$ @ $t=0$.

BEGINNING AND END OF CHANNEL, @ $x=0$, v_1^2/L, (EULERIAN APPROACH)

@ $x = L$, $2v_1^2/L$ (LAGRANGIAN APPROACH)

$x_p(t) = L(e^{v_1 t/L} - 1) \Rightarrow$ @ $x=0$, $e^{v_1 t/L} = 1$

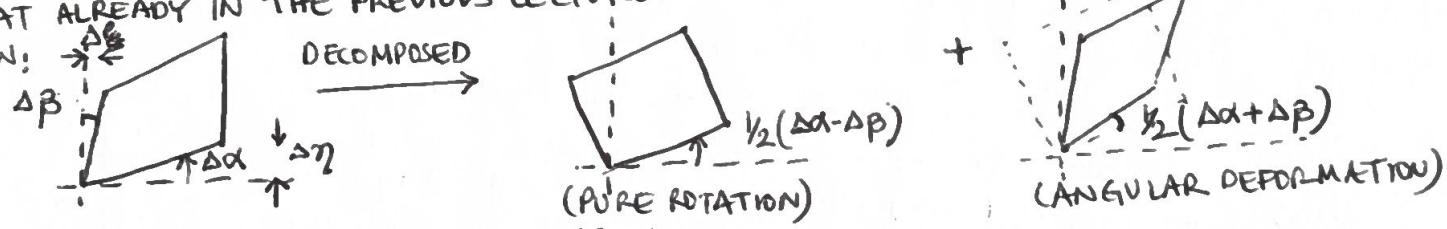
$\Rightarrow$ @ $x = L$, $e^{v_1 t/L} = 2 \Rightarrow t = L/v_1 \ln(2)$

$\therefore$ ACCELERATION $= v_1^2/L\, e^{v_1 t/L} = 2v_1^2/L$ @ OUTLET

$= v_1^2/L$ @ INLET.

} RESULT REMAINS THE SAME IRRESPECTIVE OF DESCRIPTION.

- THE TRANSLATION OF A FLUID ELEMENT IS DESCRIBED BY ITS VELOCITY AND ACCELERATION, WHICH WE LOOKED AT ALREADY IN THE PREVIOUS LECTURE.
- DERIVATION:



FOR PURE ROTATION PLUS ANGULAR DEFORMATION,

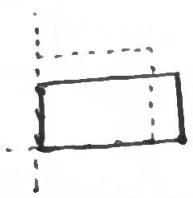
$$\Delta\xi = [u(y+\Delta y)\Delta t] - u\Delta t = \left(u + \frac{\partial u}{\partial y}\cdot\Delta y\right)\Delta t - u\Delta t = \frac{\partial u}{\partial y}\cdot\Delta y\,\Delta t = \Delta\beta\cdot\Delta y$$

$$\Delta\eta = [v(x+\Delta x)\Delta t] - v\Delta t = \left(v + \frac{\partial v}{\partial x}\cdot\Delta x\right)\cdot\Delta t - v\Delta t = \frac{\partial v}{\partial x}\cdot\Delta x\Delta t = \Delta\alpha\cdot\Delta x$$

PURE ROTATION, $\omega_z = \frac{1}{2}(\Delta\alpha - \Delta\beta) = \lim_{\Delta t\to 0}\frac{1}{2}\left(\frac{\partial v}{\partial x}\cdot\Delta t - \frac{\partial u}{\partial y}\cdot\Delta t\right)\times 1/\Delta t = \frac{1}{2}\left(\partial v/\partial x - \partial u/\partial y\right)$ → GENERAL $\omega = \frac{1}{2}\nabla\times V$

ANGULAR DEFORMATION, $\lim_{\Delta t\to 0}\frac{1}{2}\left(\frac{\partial v}{\partial x}\cdot\Delta t + \frac{\partial u}{\partial y}\cdot\Delta t\right)\times 1/\Delta t = \frac{1}{2}\left(\partial v/\partial x + \partial u/\partial y\right)$

- LINEAR DEFORMATION:



$$\{u(x+\Delta x) - u(x)\}\Delta t\Delta y + \{v(y+\Delta y) - v(y)\}\Delta t\,\Delta x = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)\cdot\Delta x\Delta y\Delta t \Rightarrow \text{VOLUME DILATION RATE} = \nabla\cdot V$$

- DERIVE THE SAME STARTING FROM VELOCITY GRADIENT TENSOR, DECOMPOSE INTO SYMMETRIC AND ANTI-SYMMETRIC COMPONENTS. SHOW THAT SYMMETRIC HAS COMPONENTS OF STRAIN RATE, ANTI-SYMMETRIC HAS COMPONENTS OF ROTATION. DIAGONAL ELEMENTS HAVE THE LINEAR DEFORMATION.
- $\omega = \frac{1}{2}\nabla\times\vec{v}$ (DEF.), $\zeta = 2\vec{\omega}$ (VORTICITY), $\Gamma = \oint_C \vec{v}\cdot d\vec{s}$ (CIRCULATION) $= \int_A (\nabla\times\vec{v})_z\, dA$ (STOKES THEOREM)

EXAMPLE: THE VELOCITY FIELD GIVEN BY $\vec{v} = Ax\hat{i} - Ay\hat{j}$ REPRESENTS FLOW IN A RECTANGULAR CORNER. EVALUATE THE CIRCULATION AROUND UNIT SQUARE WITH CORNERS (1,1), (1,2), (2,2) AND (2,1) FOR $A = 0.3$.

$\omega_z = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y)$ → ONLY THIS AS FLOW IS 2-D $\Rightarrow \omega_z = 0 \Rightarrow \Gamma = 0$.

- FOR A FREE VORTEX $v_\theta \propto 1/r$, ALTHOUGH THE TANGENTIAL COMPONENT OF THE VELOCITY IS NONZERO, THE VORTICITY IS STILL ZERO ⇒ IROTATIONAL FLOW. IN GENERAL, WE CAN ALWAYS SAY THAT ROTATION OF FLUID PARTICLES WILL OCCUR WHEN THERE ARE SHEAR STRESSES.

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- IN A STEADY FLOW, STREAMLINES, STREAKLINES & PATHLINES ARE IDENTICAL IN A FLOW FIELD.

EXAMPLE: A VELOCITY FIELD IS GIVEN BY $V = Ax\hat{i} - Ay\hat{j}$. (A) OBTAIN AN EQUATION FOR THE STREAMLINE IN THE X-Y PLANE. (B) PLOT THE STREAMLINE PASSING THROUGH THE POINT (2,8). (C) DETERMINE THE VELOCITY OF THE PARTICLE @ (2,8). (D) IF THE PARTICLE PASSING THROUGH (x_0, y_0) IS MARKED @ $t=0$, DETERMINE THE LOCATION OF THE PARTICLE @ $t=6$. (E) WHAT IS THE VELOCITY OF THIS PARTICLE @ $t=6$? (F) SHOW THAT EQUATION OF THE PARTICLE PATH IS THE SAME AS EQUATION OF STREAMLINE.

(A) STREAMLINE $dx/u = dy/v \Rightarrow dx/Ax = +dy/-Ay \Rightarrow dy/dx = -y/x \Rightarrow \ln(y) = -\ln(x) + C \Rightarrow xy = \text{CONST.}$

(B) PASSING THROUGH (2,8) ⇒ $2\times8 = \text{CONST.} = 16 \Rightarrow$ EQN. IS $xy = 16$.

(C) VELOCITY OF PARTICLE @ (2,8) = EULERIAN VELOCITY = $2A\hat{i} - 8A\hat{j}$

(D) FOR PARTICLE MOVEMENT, $dx/dt = Ax$, $dy/dt = -Ay \Rightarrow \ln(x/x_0) = At$, $\ln(y/y_0) = -At$.
PARTICLE PASSES (2,8) @ $t=0 \Rightarrow \ln(x/2) = At$, $\ln(y/8) = -At \Rightarrow x = 2e^{At}$, $y = 8e^{-At}$.

(E) @ $t=6$, $dx/dt = 2Ae^{At}$, $dy/dt = -8Ae^{-At} \Rightarrow$ @ $t=6$, $u = 2Ae^{6A}$, $v = -8Ae^{-6A}$.

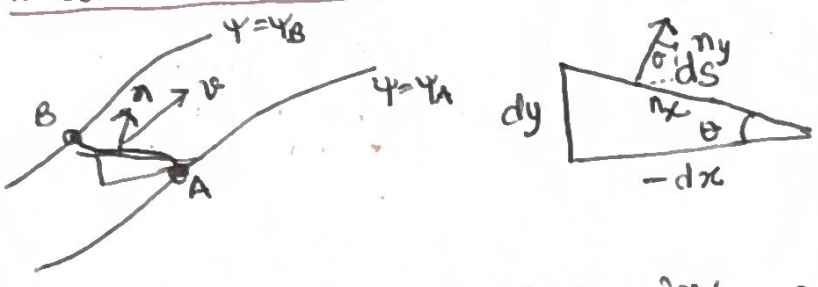
(F) FROM EQNS., $xy = 2e^{At}\cdot 8e^{-At} \Rightarrow xy = 16 =$ (SAME AS PART (B)).

FROM PARTICLE PATH EQN., $\ln(x/x_0) = A(t-t_0)$, $\ln(y/y_0) = -A(t-t_0)$

THE PARTICLES ON THE STREAKLINE HAVE PASSED THROUGH (x_0, y_0) FOR SOME $0 \le \tau \le t$

$\Rightarrow \ln(x/x_0) = A(t-\tau)$, $\ln(y/y_0) = -A(t-\tau) \Rightarrow xy = \text{CONST.}$ (EQN. FOR STREAKLINE).

- 2D-FLOW ⇒ $\partial u/\partial x + \partial v/\partial y = 0 \Rightarrow u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x \Rightarrow$ AUTOMATICALLY SATISFIES INCOMPRESSIBILITY.
- ALONG A STREAMLINE, $dx/u = dy/v \Rightarrow u\,dy - v\,dx = 0 \Rightarrow \partial\psi/\partial y \cdot dy + \partial\psi/\partial x \cdot dx = 0 \Rightarrow d\psi = 0 \Rightarrow$ STREAMFUNCTION IS CONSTANT ALONG A STREAMLINE.



$$Q_{AB} = \int_{A-B} (n\cdot v)\,dS = \int_A^B (n_x u + n_y v)\,ds$$

FROM GEOMETRY, $\sin\theta = dy/ds = n_x/n \Rightarrow dy = n_x\,ds$
$\cos\theta = -dx/ds = n_y/n \Rightarrow dx = -n_y\,ds$

$$\Rightarrow Q_{AB} = \int_A^B (u\,dy - v\,dx) = \int_\psi^B d\psi = \psi_B - \psi_A.$$

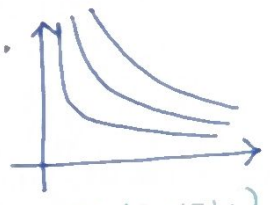
FOR IRROTATIONAL, $\omega_z = 0 \Rightarrow \nabla^2\psi = 0$ (FOR VORTICITY-STREAMFUNCTION)

- FROM FORMULA, $\omega = \omega_z = -\partial u/\partial y + \partial v/\partial x \Rightarrow \omega_z = -\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2$
- SIDENOTE: 2D STEADY COMPRESSIBLE FLOW, $\partial/\partial x(\rho u) + \partial/\partial y(\rho v) = 0$; $\rho u = \partial\psi/\partial x$, $\rho v = -\partial\psi/\partial x \Rightarrow \psi_B - \psi_A$ GIVES MASS FLOW RATE PER UNIT DEPTH.

EXAMPLE: $V = Ax\hat{i} - Ay\hat{j}$. DETERMINE STREAMFUNCTION THAT WILL GIVE THIS VELOCITY FIELD. PLOT AND INTERPRET THE STREAMLINE PATTERN IN THE FIRST AND SECOND QUADRANTS.

$Ax = \partial\psi/\partial y$, $-Ay = -\partial\psi/\partial x \Rightarrow \psi = Axy + G_1(x)$, $\psi = +Axy + G_2(y) \Rightarrow \psi = Axy + \text{CONSTT.}$

WHEN $\psi \equiv$ CONSTANT, THAT CORRESPONDS TO A STREAMLINE.



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- CYLINDRICAL COORDINATES: $v_r = -\partial\phi/\partial r$, $v_\theta = -\frac{1}{r}\partial\phi/\partial\theta$, $v_z = -\partial\phi/\partial z$. (EMPHASIZE THAT ϕ EXISTS FOR 3D AS WELL)
- INCOMPRESIBLE + IRROTATIONAL + 2D ⇒ $\nabla^2\phi = \nabla^2\psi = 0$ (ANY ψ OR ϕ THAT SATISFIES LAPLACE EQN. REPRESENTS A POSSIBLE 2D INCOMPRESSIBLE IRROTATIONAL FLOW FIELD)
- ALONG A STREAMLINE, $\psi = \text{CONST.} \Rightarrow d\psi = \partial\psi/\partial x\cdot dx + \partial\psi/\partial y\cdot dy = 0 \Rightarrow -v\,dx + u\,dy = 0 \Rightarrow dy/dx = v/u$ → -1 PRODUCT (ORTHOGONAL)
  ALONG EQUIPOTENTIAL, $\phi = \text{CONST.} \Rightarrow d\phi = \partial\phi/\partial x\cdot dx + \partial\phi/\partial y\cdot dy = 0 \Rightarrow -u\,dx - v\,dy = 0 \Rightarrow dy/dx = -u/v$.

EXAMPLE: THE STREAMFUNCTION OF A FLOW FIELD IS $\psi = Ax^3 + B(xy^2 + x^2 - y^2)$, WHERE ψ, x, y, A, B ARE ALL DIMENSIONLESS. FIND THE RELATION B/W A AND B FOR THIS TO BE AN IRROTATIONAL FLOW, AND FIND POTENTIAL.

$\nabla^2\psi = 6Ax + B(2) + 2Bx - 2 = 6Ax + 2Bx + 2B - 2 = 0$ (FOR IRROTATIONAL) $\Rightarrow x(6A + 2B) + 2B - 2 = 0$
$\Rightarrow B = 1$, $A = -1/3$

$u = \partial\psi/\partial y = 2Bxy - 2By$, $v = -\partial\psi/\partial x = -3Ax^2 + By^2 + 2Bx$, $u = -\partial\phi/\partial x \Rightarrow -\partial\phi/\partial x = 2Bxy - 2By \Rightarrow \phi = -Bx^2y + 2Bxy + f(y)$
$v = -\partial\phi/\partial y \Rightarrow -\partial\phi/\partial y = -3Ax^2 + By^2 + 2Bx \Rightarrow \phi = -3Ax^2y + \frac{B}{3}y^3 + 2Bxy + g(x)$

- SINCE LAPLACE EQN. IS A LINEAR EQUATION, SOLUTIONS CAN BE SUPERIMPOSED.
- SINCE WE KNOW THAT THERE IS NO FLOW ACROSS A STREAMLINE, IT CAN BE INTERPRETED AS A SOLID SURFACE.

EXAMPLE: A FLOW FIELD IS FORMED BY COMBINING A UNIFORM FLOW IN THE POSITIVE-X DIRECTION, WITH $U = 10$ m/s, AND A COUNTER-CLOCKWISE VORTEX WITH STRENGTH $K = 16\pi$ m²/s LOCATED AT THE ORIGIN. OBTAIN THE STREAM FUNCTION, VELOCITY POTENTIAL AND VELOCITY FIELD FOR THE COMBINED FLOW. LOCATE THE STAGNATION POINTS OF THE FLOW. PLOT THE STREAMLINES AND POTENTIAL LINES.

$\psi = Ur\sin\theta - \frac{K}{2\pi}\ln(r)$, $\phi = -Ur\cos\theta - \frac{K}{2\pi}\theta$ (FROM FORMULA) $\Rightarrow \psi = 10r\sin(\theta) - 8\ln(r)$, $\phi = -10r\cos\theta - 8\theta$

$v_r = -\frac{\partial\phi}{\partial r} = +10\cos\theta$, $v_\theta = -\frac{1}{r}\frac{\partial\phi}{\partial\theta} = -\frac{1}{r}[10r\sin\theta - 8] = -10\sin\theta + 8/r$, @ STAGNATION, $V = 0 \Rightarrow 10\cos\theta\,\hat{e}_r + (8/r - 10\sin\theta)\,\hat{e}_\theta = 0$

$\Rightarrow \cos\theta = 0 \Rightarrow \theta = (2n+1)\pi/2$
$8/r = 10\sin\theta \Rightarrow r = 4/5$.